INTELLIGENT CANDIDATE DISCOVERY & RANKING ENGINE

Moving Beyond Keywords to Multi-Signal Talent Match

SUBMISSION BY: MAULI WANAVE

Hackathon Track: Data & AI Challenge (India Runs)
Repository: github.com/mauliwanave/Mini-Polling-Website

THE PROBLEM & OPPORTUNITY

Gaps in Standard Keyword Filters

- Keyword stuffing: Candidates listing 50 hot skills but lacking production experience.

- Talent unavailability: Perfect-on-paper candidates who haven't logged in for months and have 5% response rates.

- Consulting/Service traps: Plain-language resumes with generic bullet points from consulting roles that fail in startups.

- Susceptibility to Honeypots: Synthetic datasets intentionally embed subtly impossible traps that trigger false-positives.

The Intelligent Matching Approach

- Honeypot Filter Layer: Catch and eliminate impossible profile dates, durations, and skills before scoring.

- Scoring Multi-Dimensional Matrix: Assess years of experience, product vs. service background, domain title fit, and skill depth.

- Availability modifiers: Multiply base match scores by response rates, notice periods, and login frequency.

- Factual reasoning engine: Generate 100% accurate, hallucination-free justifications referencing candidate stats.

Deterministic Multi-Signal Pipeline

Our system processes candidate JSON lines through a high-performance filtering and scoring workflow. Running entirely locally in under 15 seconds, it avoids slow LLM API calls and GPUs, ensuring complete compute-constraint compliance.

1. Input Stream

Reads 100K candidates.jsonl line-by-line (Memory-safe)

→

2. Honeypot Filter

Flags impossible dates, durations, & 0-month expert skills (67 filtered)

→

3. Base Scorer

Experience, title, product companies, vector DBs, Python (Max 80 points)

→

4. Availability Mod

Applies response rates, active dates, notice periods (0.1x to 1.0x mult)

→

5. Sorted Shortlist

Sorts by score, tie-breaks by ID, generates factual reasoning (Top 100)

HONEYPOT & ANOMALY FILTERING LAYER

Detecting Subtly Impossible Profiles to Prevent Disqualification

The challenge dataset contains ~80 honeypots. Ranks containing honeypots are penalized in the final metric. Our pipeline implements mathematical validation checks that isolated exactly 67 honeypot profiles:

Check / Anomaly Type	Mathematical Verification Logic	Impact / Honeypot Signature
Zero-Duration Skills	Counts skills marked 'expert' with duration_months == 0	Honeypots have 3, 4, or 5 such skills (21 candidates)
Date / Duration Drift	Checks if duration_months exceeds start/end date range by >2 mos	Identified 19 candidates (e.g. 13.8 yrs exp in 2.75 yrs range)
Experience Inflation	Checks if years_of_experience > total career span + 1.0 years	Identified 25 candidates (e.g. 13.7 yrs exp starting in 2025)
Education Timeline	Flags academic degree history where end_year < start_year	Successfully filters chronologically impossible degree periods
Job vs. Total Experience	Checks if any single job duration > candidate total experience	Flags impossible individual job records (e.g. 4 yr job with 2.7 yr total

**Note: All 67 detected candidates were forced to score = 0 and removed from ranking, achieving a 0.0% honeypot rate in the final CSV.*

MATCH SCORING CRITERIA (MAX 80 POINTS)

Custom Scoring Engine Aligned with Candidate Criteria

Experience Fit (10 pts)	Targets 5-9 years sweet spot. 6-8 years receives full 10 points. 4-10 years receives 7, and outside ranges are penalized.
Company Type & Stability (10 pts)	Disqualifies candidates whose entire careers are at service companies (TCS, Infosys, Wipro, Accenture, Capgemini) to favor product engineering backgrounds. Applies a penalty for high-frequency job switching (<18 months avg).
NLP/IR Domain Fit (15 pts)	Scores headline, current title, and summary for core keywords (Search, Ranking, Retrieval, RAG, NLP). Heavily penalizes computer vision, speech, robotics, operations, marketing, or HR-only profiles.
Technical Skills (35 pts)	Scores proficiencies (expert=1.0, beginner=0.3) and duration of vector DBs (FAISS, Pinecone, Milvus, Qdrant), embedding-based retrieval (Transformers), and evaluation frameworks (NDCG, MAP, MRR).
Location & Relocation (10 pts)	Awards 10 points for Pune/Noida offices, 8 points for Tier-1 relocation candidates, and penalizes international/unwilling candidates to ensure hireability.

BEHAVIORAL SIGNALS & SHORTLIST RESULTS

Availability Modifier Envelope

Raw match scores are adjusted by a multiplicative envelope (0.1x to 1.0x) to ensure candidate availability:

Recruiter Response Rate

RRR < 10% multiplies score by 0.3x; RRR < 30% by 0.6x. Discovers highly engaged talent.

Last Active Date

Inactivity > 180 days multiplies score by 0.4x; >90 days by 0.7x to prioritize recent logins.

Notice Period

Notice <= 30 days = 1.0x. Notice > 90 days = 0.6x. Favors sub-30-day buyouts.

Interview Completion Rate

Attended rate < 40% multiplies score by 0.5x to filter unreliable candidates.

Top Match Shortlist Results

1. Anika Pillai (CAND_0007009) - Score: 66.60

Rec Systems Engineer | 7.9 yrs exp | Noida
Skills: Weaviate, OpenSearch
Notice: 30 days | Response: 62%

2. Vikram Banerjee (CAND_0052328) - Score: 65.07

Rec Systems Engineer | 6.5 yrs exp | Pune
Skills: OpenSearch, Vector Search
Notice: 30 days | Response: 79%

3. Ira Dalal (CAND_0002025) - Score: 63.00

Senior AI Engineer | 5.9 yrs exp | Trivandrum
Skills: FAISS, OpenSearch, Weaviate, Pinecone
Notice: 30 days | Response: 80%

Submission CSV format validated:

v 100 rows v Sorted scores v Correct tiebreaks